LAB WORKSHEET

ACL Lab

Filter traffic on a Cisco router with access control lists: wildcard masks, standard vs extended ACLs, placement and direction.

Key facts

- **Standard** ACLs (1–99, 1300–1999) match the **source** only. Place them close to the **destination**.
- **Extended** ACLs (100–199, 2000–2699) match protocol, source, destination and port. Place them close to the **source**.
- Wildcard = 255.255.255.255 − subnet mask (/24 → `0.0.0.255`). `host x` = `x 0.0.0.0`; `any` = `0.0.0.0 255.255.255.255`.
- ACLs are read top-down and stop at the **first match**, with an invisible **implicit deny** at the end. You can have one ACL per interface, per direction, per protocol.
- Router ACLs are **not stateful**: replies are checked as well, so use `established` for TCP replies.

Key commands

```
access-list 10 deny 192.168.20.0 0.0.0.255
access-list 10 permit any
interface g0/0
 ip access-group 10 out
ip access-list extended STUDENT_POLICY
 permit tcp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq www
 deny tcp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 445
 permit ip any any
interface g0/1
 ip access-group STUDENT_POLICY in
line vty 0 4
 access-class 5 in
 transport input ssh
show access-lists        show ip interface g0/1        no 20
```

Tabs

1. **Learn:** reference, the wildcard-mask calculator and a scored quiz.
2. **ACL practice:** wildcard drill, build-a-statement (12 scenarios) and order puzzles.
3. **Configure the router:** topology plus the IOS console. Test packets show where they are checked (in/out) and which line matched.

Guided task

Log in (enable password `class`) → check the interfaces → standard ACL 10 out on G0/0 → test → named extended STUDENT_POLICY in on G0/1 → test → add `permit ip any any` → lock the VTY lines to Admin-PC → `show access-lists` → save.

Independent task

Hollinbrook Academy (fictional): meet 8 requirements with no commands given, then evidence them with the test log.